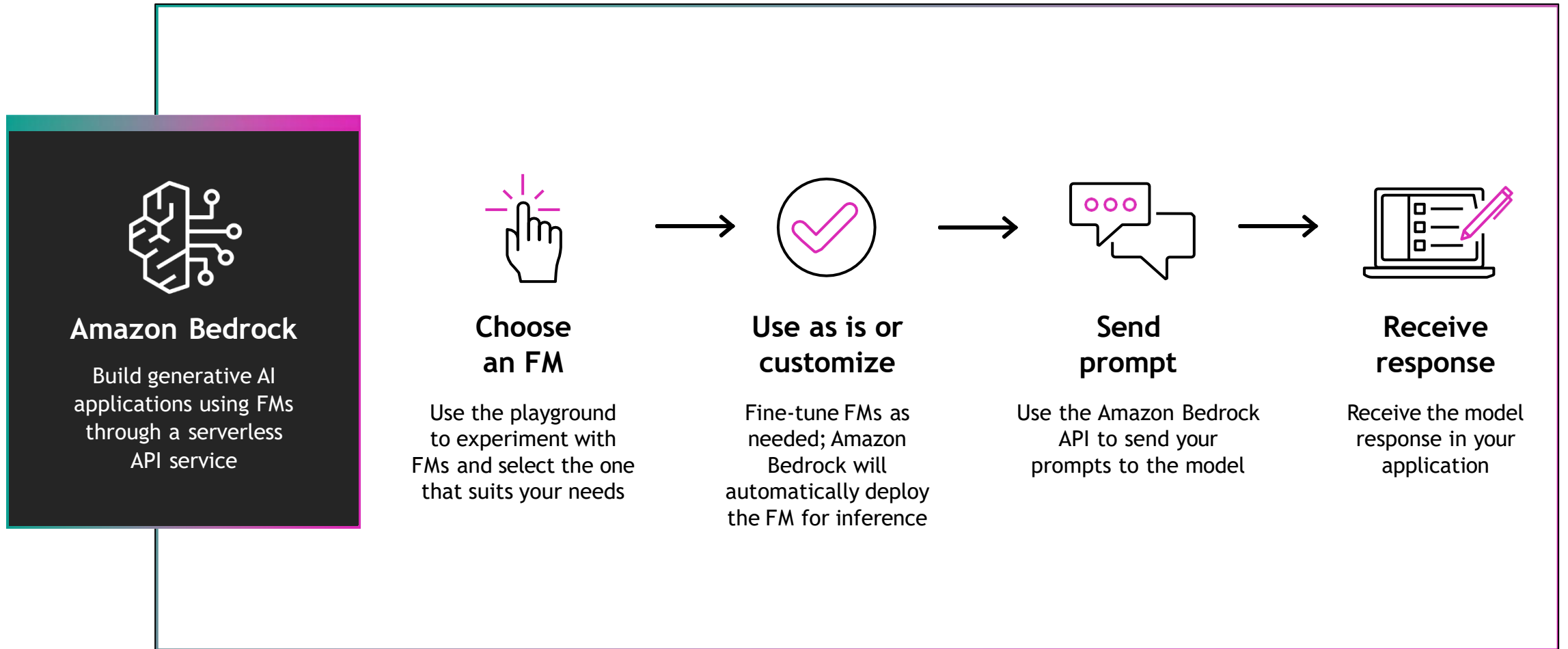


Amazon Bedrock

AWS AI Service

- AWS Service for AI Implementation
- The easiest way to build and scale generative AI applications with foundation models
 - Choice of leading FMs via single API
 - Model customization
 - Retrieval Augmented Generation (RAG)
 - Agents that execute multistep tasks
 - Security, privacy, and safety

How Amazon Bedrock works



Feature Overview and Console Navigation



Discover

Menu	Description
Overview	Dashboard showing Bedrock capabilities, latest updates, quick links, and status of services.
Model catalog	View all available foundation models (FMs) from Amazon and third parties (Anthropic Claude, Stability, Meta, Mistral, Cohere, etc.). Includes metadata, pricing, and inference parameters.
API keys	Manage API keys for using Bedrock APIs via SDK or CLI.
Chat / Text playground	Interactive chat UI for text-based models (Claude, Titan Text, Mistral, etc.).
Image / Video playground	Interface for image generation or video understanding models (e.g. Stability AI).
Watermark detection	Tool to check whether an image includes an AI-generated watermark (for responsible AI and compliance).

Feature Overview and Console Navigation



Inference

Menu	Description
Cross-region inference	Run inference on models hosted in other AWS regions for latency or availability reasons.
Batch inference	Submit multiple inference requests in bulk — useful for large-scale generation tasks.
Provisioned Throughput	Reserve dedicated capacity for predictable, low-latency inference.
Custom model on-demand (New)	Upload or deploy your own model artifacts for on-demand inference.

Feature Overview and Console Navigation



Tune

Menu	Description
Custom models	Fine-tune or train Bedrock models with your own labeled data.
Prompt router models	Configure routing logic to direct prompts to different models based on input characteristics or performance metrics.
Imported models	Import and register models from outside Bedrock (e.g., SageMaker or your own Hugging Face model).
Marketplace model deployment	Deploy models sourced from AWS Marketplace directly within Bedrock.

Feature Overview and Console Navigation



Build

Menu	Description
Agents	Create intelligent agents that use foundation models to reason, plan, and call APIs or knowledge bases.
Flows	Build multi-step GenAI workflows visually — combining model calls, Lambda functions, conditional logic, and data transformations.
Knowledge Bases	Add Retrieval-Augmented Generation (RAG) capability — connect to your data sources and use embeddings/vector stores for contextual responses.
Automated Reasoning	Use formal reasoning tools to verify logic or constraints in agentic workflows.
Guardrails	Define rules for safe, compliant, and responsible model behavior (e.g., profanity filters, topic restrictions).
Prompt Management	Manage, version, and reuse prompts across flows, agents, and evaluations.
Data Automation	Automate data ingestion, cleaning, or preparation pipelines for training and fine-tuning.
AgentCore	Framework to deploy, scale, and monitor AI agents at enterprise level with Bedrock's orchestration primitives.

Feature Overview and Console Navigation



Assess

Menu	Description
Evaluations	Evaluate and compare model outputs for quality, accuracy, safety, latency, and cost metrics. Useful for prompt tuning and model selection.

Configure and learn

Menu	Description
Settings	General configuration (region, permissions, environment variables, etc.).
Model access	Control which foundation models your AWS account or users can access.
User guide	Direct link to the official Bedrock documentation.
Bedrock Service Terms	Legal and compliance information for Bedrock service usage.

Model Catalog

AWS AI Discover Service

Amazon Bedrock Models



AI21labs

amazon

ANTHROPIC

cohere

Meta

MISTRAL AI

stability.ai

Contextual answers, summarization, paraphrasing

Text summarization, generation, Q&A, search, image generation

Summarization, complex reasoning, writing, coding

Text generation, search, classification

Q&A and reading comprehension

Text summarization, Q&A, text classification, text completion, code generation

High-quality images and art

Jurassic-2 Ultra

Amazon Titan Text Lite Amazon

Claude 3 Opus

Command Command

Llama 3 8B

Mistral Large

Stable Diffusion XL1.0

Jurassic-2 Mid

Titan Text Express Amazon

Claude 3 Sonnet

Light Embed English

Llama 3 70B

Mistral 7B

Stable Diffusion XL 0.8

Titan Text Premier

Claude 3 Haiku

Embed Multilingual

Llama 2 13B

Mixtral 8x7B

Amazon Titan Text Embeddings

Claude 2.1

Command R+

Llama 2 70B

Amazon Titan Text Embeddings V2

Claude 2 Claude

Command R

Amazon Titan Multimodal Embeddings

Instant

Amazon Titan Image Generator

Model Catalog



The screenshot shows the AWS Bedrock Model Catalog interface. The left sidebar contains navigation links for Amazon Bedrock, Discover, Test, Infer, Tune, and Build. The main content area displays the 'Model catalog (243)' with a description and a 'Filters' section. The filters applied are 'Model collection', 'Providers', 'Modality', 'Serverless model access status = Access granted', and 'Marketplace deployments'. A 'Spotlight' search bar shows 'Filter for a model' with 36 matches. The models listed are:

Model Name	Access Status	Capabilities	Deployment Type
Llama 4 Maverick 17...	Access granted	Fine Tuning, Text summarization, Code generation, Function calling, Advanced...	Serverless
Llama 4 Scout 17B...	Access granted	Fine Tuning, Text summarization, Code generation, Function calling, Advanced...	Serverless
Llama 3.3 70B Instruct	Access granted	Tool Use, Code Generation, Advanced reasoning and decision making, Steerability	Serverless
Llama 3.2 11B Vision...	Access granted	Chat optimized, Code generation, Complex reasoning analysis, Conversation, Documen...	Serverless
Llama 3.2 1B Instruct	Access granted	Agents, Chat optimized, Code generation, Complex reasoning analysis, Conversation,...	Serverless
Llama 3.2 3B Instruct	Access granted	Agents, Chat optimized, Code generation, Complex reasoning analysis, Conversation,...	Serverless

Model Request



us-east-1.console.aws.amazon.com/bedrock/home?region=us-east-1#/modelaccess

Amazon Bedrock > Model access

Model access page retiring soon
Amazon Bedrock will simplify access to all serverless foundation models, and any new models, by automatically enabling them for every AWS account in commercial AWS regions, eliminating the need to manually activate access through the Bedrock console. Account administrators retain full control over model access through [IAM policies](#) and [Service Control Policies \(SCPs\)](#) to restrict model access as needed.
Note: Selected new models launched before model access is simplified will be auto-enabled, so they won't appear on this page; you can still manage their access via [IAM](#) / [SCPs](#). For a complete list of supported models, refer to our [documentation](#).

What is Model access?
To use Bedrock serverless models, account users with the correct [IAM Permissions](#) must enable access to available Bedrock foundation models (FMs). View all [Bedrock Model Terms](#) for Bedrock FMs.
[Modify model access](#)
Visit [Amazon Bedrock Quotas](#) for a quick guide to the default quotas and limits that apply to Amazon Bedrock.

Base models (62)
Not seeing a model you're interested in? Check out all supported models by region [here](#).
 Group by provider

Models	Access status	Modality	EULA
AI21 Labs (2)	0/2 access granted		
Jamba 1.5 Large	Request failed	Text	EULA
Jamba 1.5 Mini	Request failed	Text	EULA

CloudShell Feedback

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Model Request



us-east-1.console.aws.amazon.com/bedrock/home?region=us-east-1#/modelaccess

Search [Alt+S] United States (N. Virginia) stalwartlabs (5901-8367-9875) SL_User1_AR

Amazon Bedrock > Model access > Manage model access

Model access page retiring soon
Amazon Bedrock will simplify access to all serverless foundation models, and any new models, by automatically enabling them for every AWS account in commercial AWS regions, eliminating the need to manually activate access through the Bedrock console. Account administrators retain full control over model access through [IAM policies](#) and [Service Control Policies \(SCPs\)](#) to restrict model access as needed.
Note: Selected new models launched before model access is simplified will be auto-enabled, so they won't appear on this page; you can still manage their access via [IAM](#)/[SCPs](#). For a complete list of supported models, refer to our [documentation](#).

Step 1: Edit model access
Step 2: Review and submit

Edit model access

Base models (12/62) [Collapse all](#)

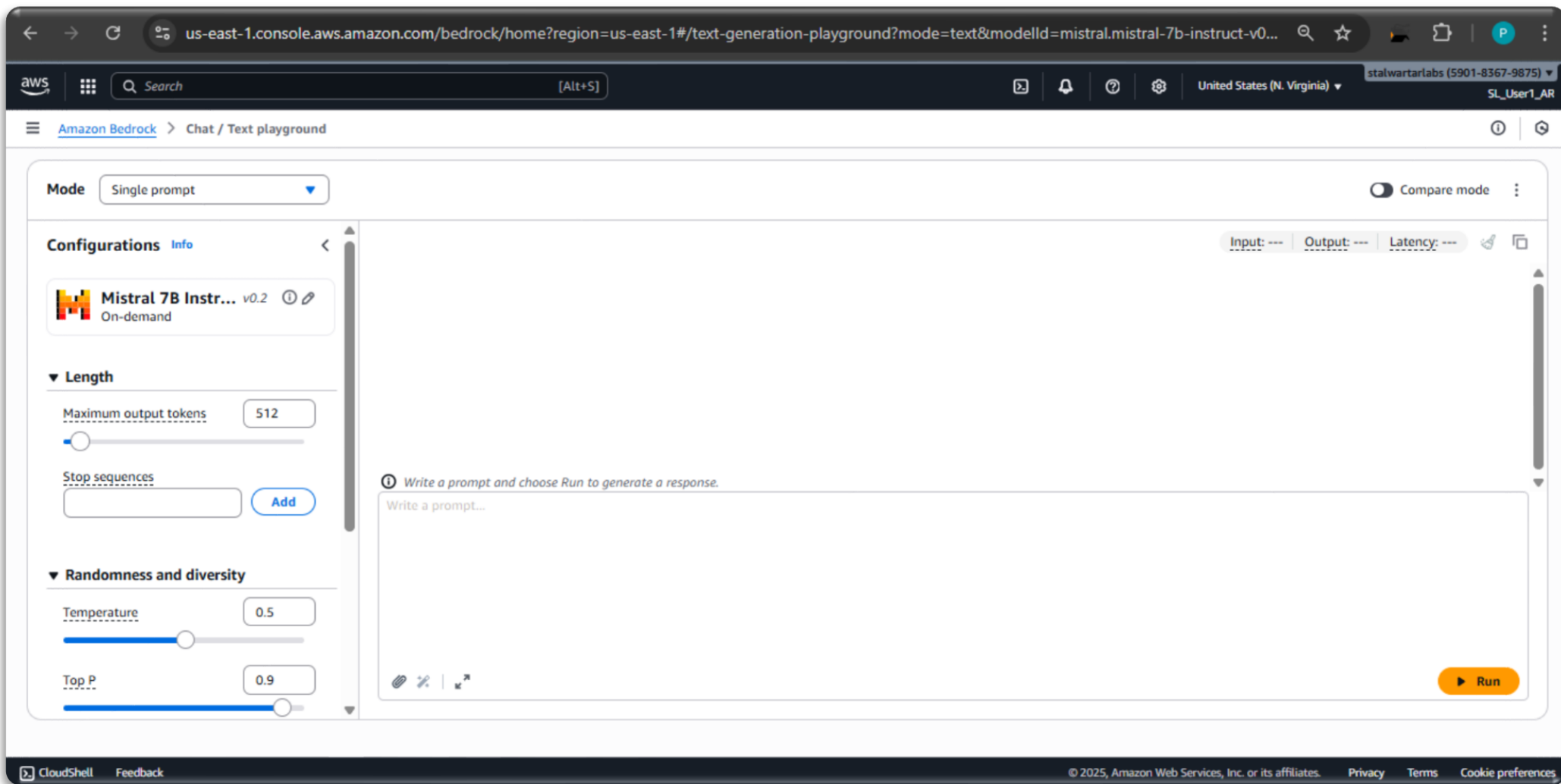
Not seeing a model you're interested in? Check out all supported models by region [here](#).

Find model Group by provider

Models	Access status	Modality	EULA
☐ AI21 Labs (2)	0/2 access granted		
☐ Jamba 1.5 Large	Request failed	Text	EULA
☐ Jamba 1.5 Mini	Request failed	Text	EULA
☒ Amazon (14)	14/14 access granted		
☒ Nova Pro	Access granted	Text	EULA
☒ Titan Text G1 - Express	Access granted	Text	EULA

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Playground



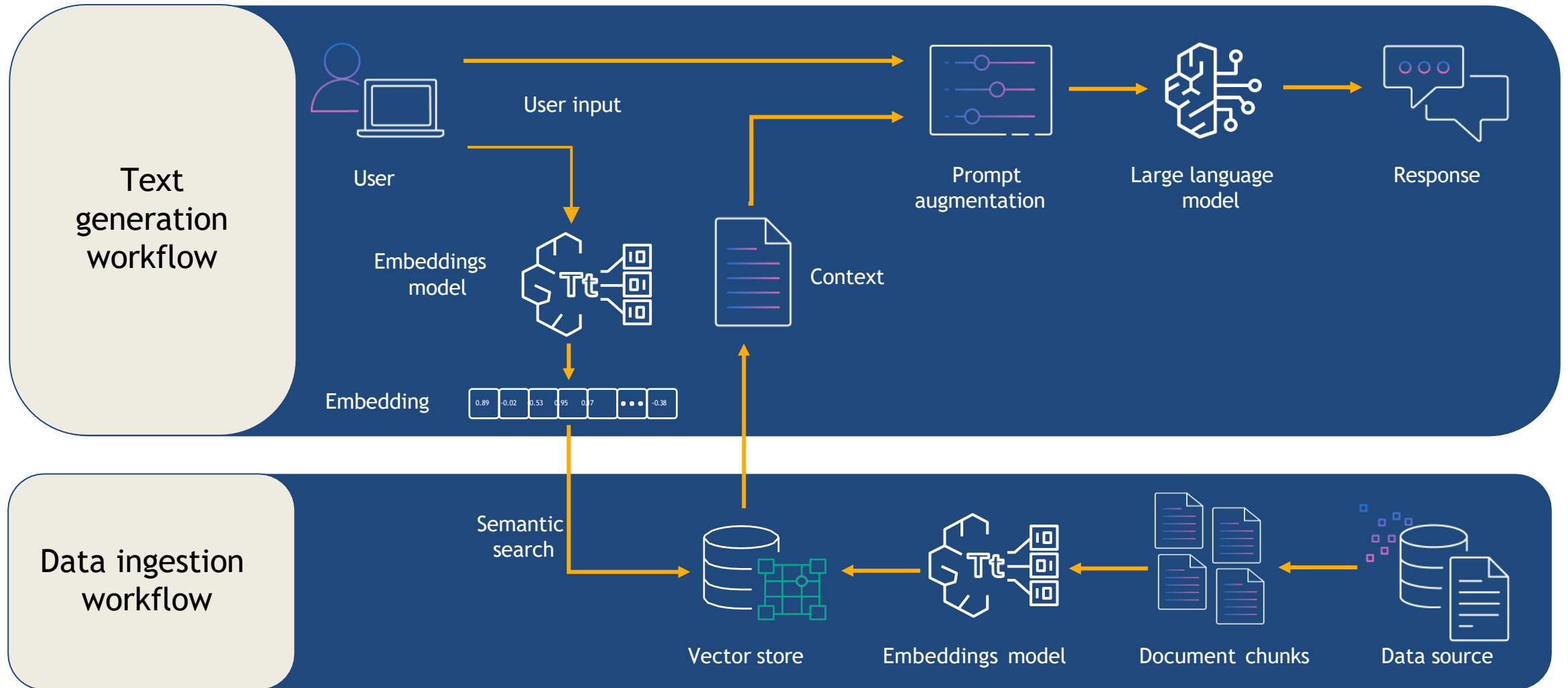
Knowledge Base

AWS AI Build Service

- A Knowledge Base is a fully managed **Retrieval-Augmented Generation (RAG)** service that allows a foundation model to access and use your own data to give contextually relevant and accurate answers - without retraining the model.

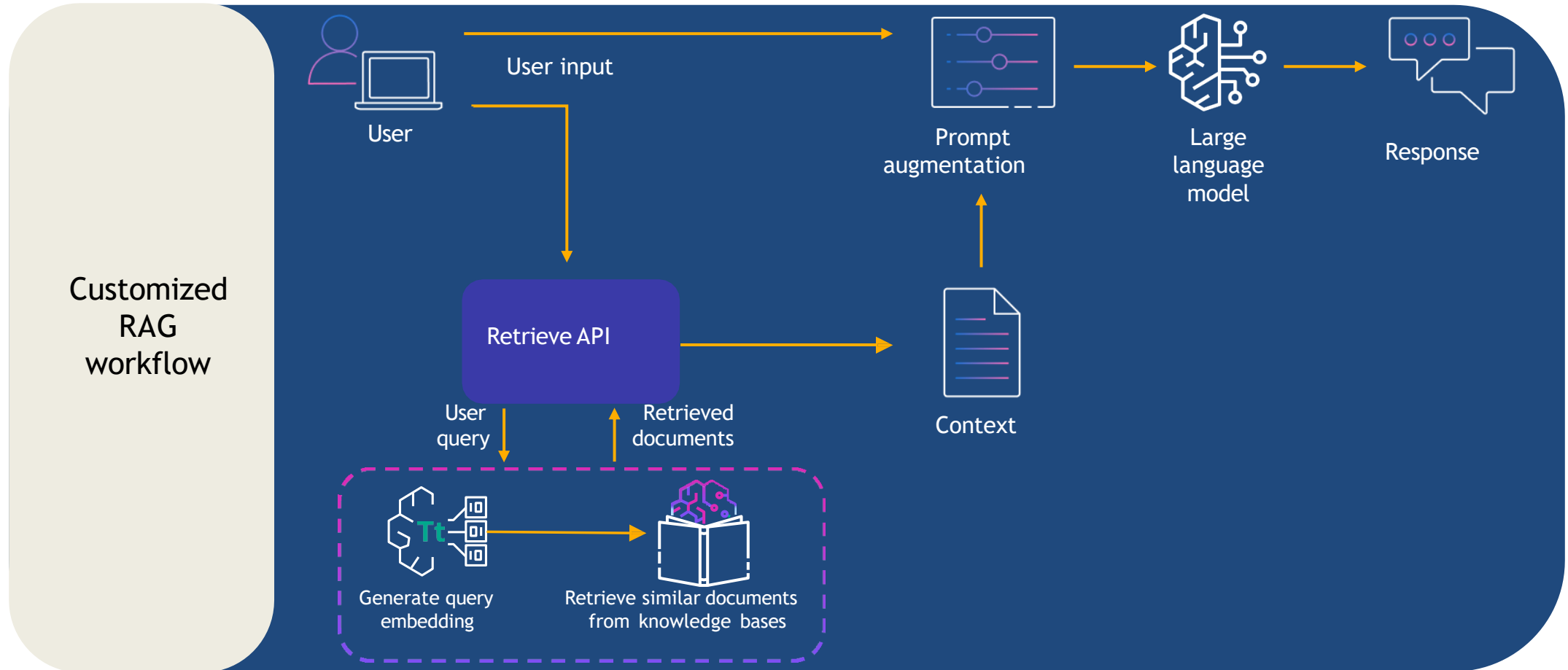
- RAG provides your model with additional information that you can provide the FM as context; this improves the relevance of the response

RAG in action



Customize RAG workflows using Retrieve API

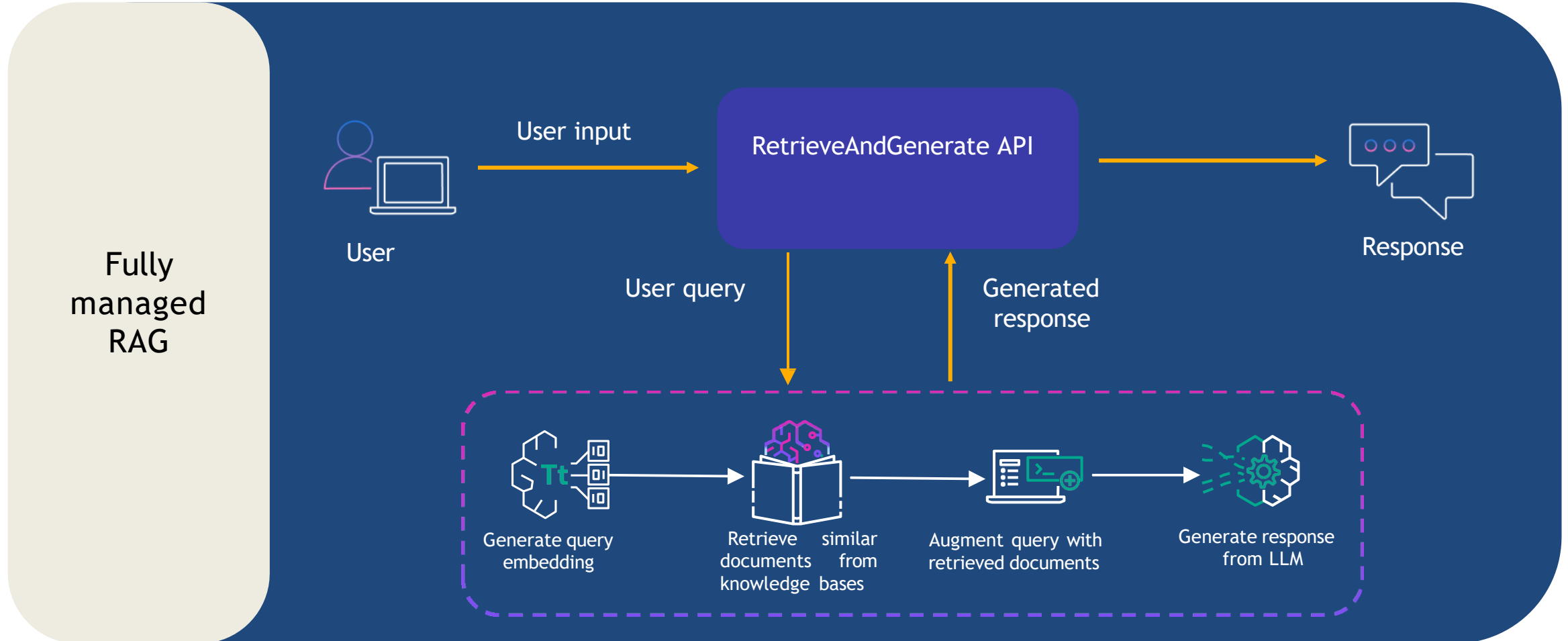
Knowledge bases for amazon bedrock



RetrieveAndGenerate API



Knowledge bases for amazon bedrock

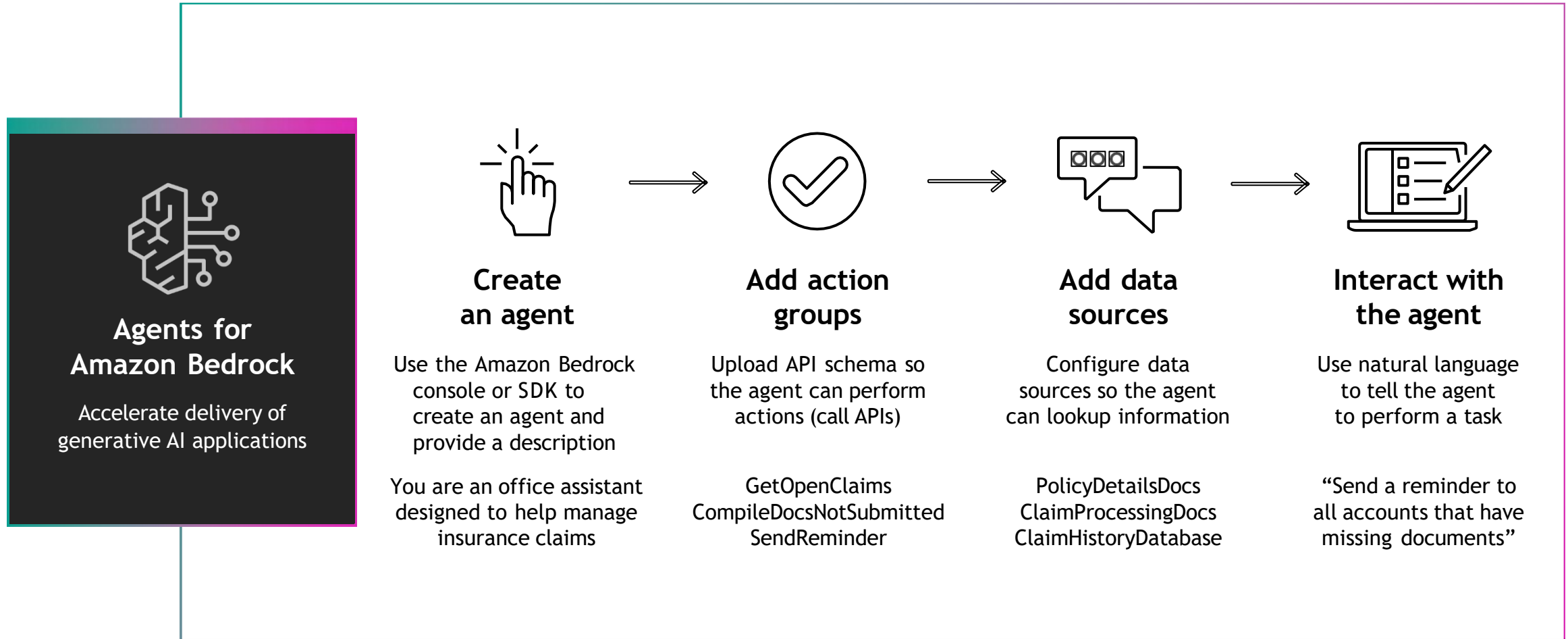


Agents

AWS AI Build Service

- An Agent is a managed **orchestration** component that enables a foundation model (FM) to plan, reason, and act - meaning it can break down user requests into steps, call external APIs or AWS services, retrieve knowledge, and generate final responses autonomously.

How it works: Agents



Model Evaluation

AWS AI Assess Service

Model Evaluation



Model evaluation is a service within Amazon Bedrock that allows users to assess and compare the performance of generative AI models.

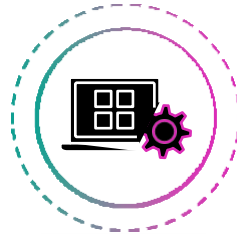
Model Evaluation



Playground

Evaluate directly in the playground as you try out different models; compare across cost latency and usage dimensions

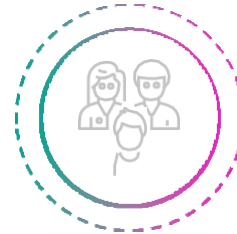
Use playground as you narrow down on the use case and identify the FM



Programmatic

Evaluate as part of your application development lifecycle or model customization; metrics include accuracy, toxicity, and robustness

Use programmatic evaluation as you iterate on the use case or the model



Human-in-the-loop Bring your own team

Use the framework offered to organize your team for your evaluation; better suited for subjective criteria such as brand voice, clarity, and tone

Bring your own team as you start testing your first prototype or get ready for pilot



Human-in-the-loop AWS Managed team

Use AWS expertise for an expert evaluation, in case you don't have the budget or expertise in-house

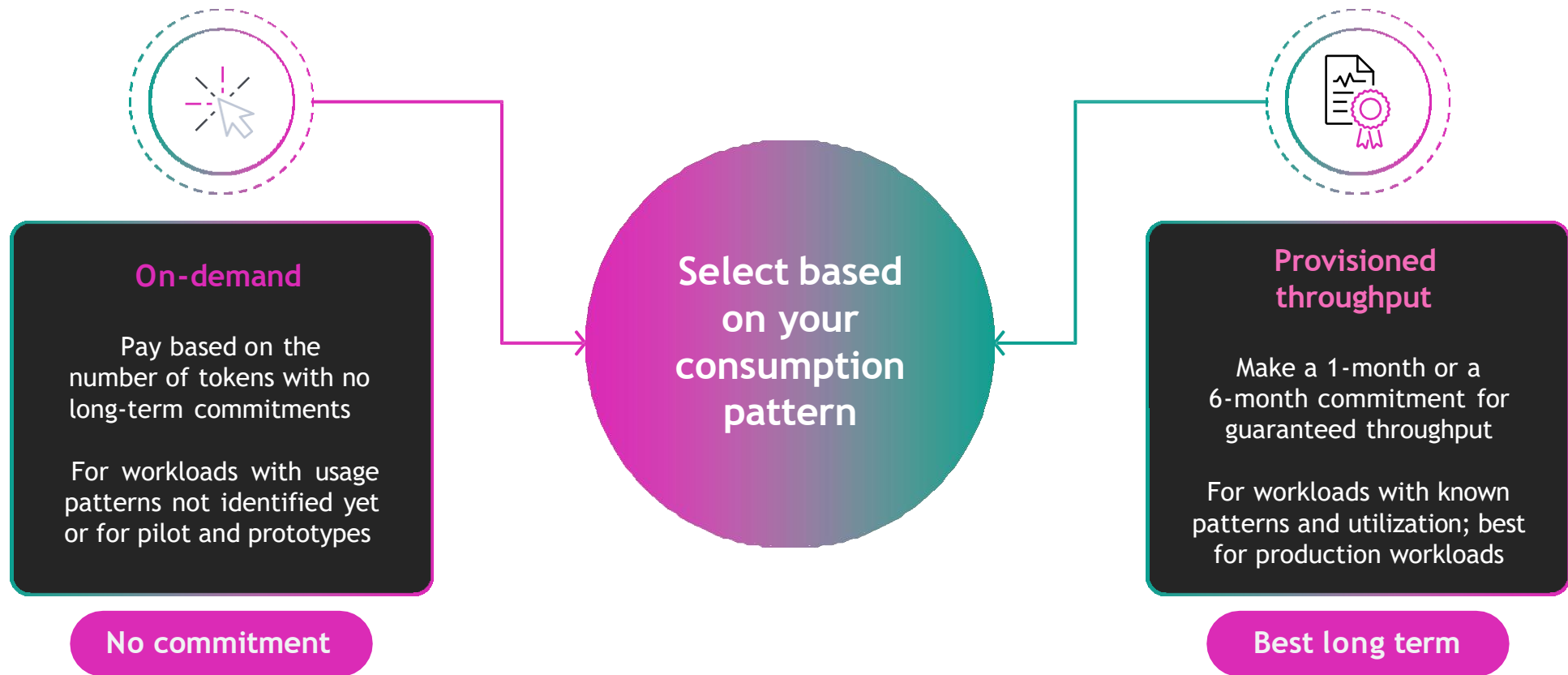
AWS-managed team as you get ready for production launch of your application



Integration

AWS AI Inference Service

Consumption choice



Provisioned throughput



Allocate model units

Engage with AWS to allocate model units for a specific model

Each model unit provides a predetermined level of guaranteed throughput



Purchase throughput

Once allocated, purchase throughput directly from the console

Guaranteed throughput for the term of commitment—option to renew automatically



Amazon Bedrock > Provisioned throughput > Purchase provisioned throughput

Purchase provisioned throughput [info](#)

Provisioned throughput details [info](#)

Provisioned throughput name

Name can have up to 40 characters, and it must be unique. Valid characters A-Z, a-z, 0-9, and - (hyphen).

Select model

Tags - optional

Model units & commitment term [info](#)
Select model units & commitment term to purchase Provisioned throughput. To estimate cost use [MU Estimator](#).

Model units
Please request the model units here before purchasing provisioned throughput. [AWS support center](#)

Select commitment term
Commitment terms locks the purchase for the selected duration.

Estimated purchase summary
To view the provisioned throughput pricing please visit: [Pricing information](#)

Estimated hourly cost	Estimated daily cost	Estimated monthly cost
—	—	—

Edits to model and model units will be restricted
Once provisioned throughput is purchased, model units cannot be updated and the model can only be updated to another model with the same lineage. [Learn more](#)

Guardrails

AWS AI Build Service

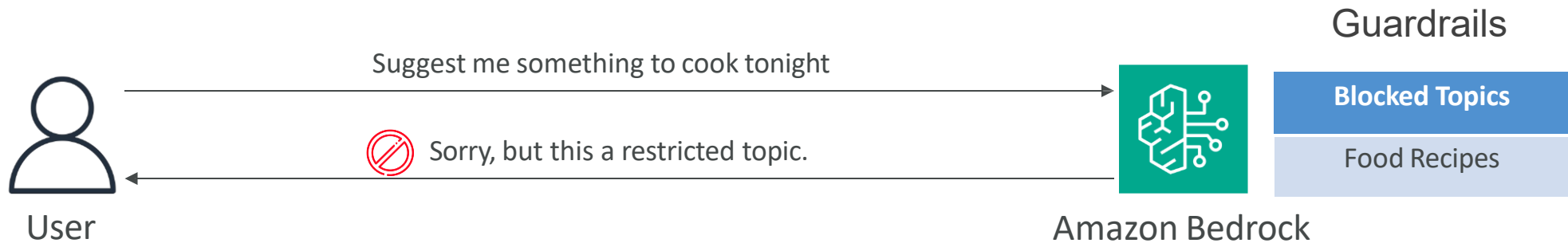
Implement application-specific safeguards based on your use cases and responsible AI policies

- Apply guardrails to multiple foundation models and agents for Amazon Bedrock
- Configure harmful content filtering based on your responsible AI policies
- Define and disallow denied topics with short natural language descriptions
- Redact sensitive PII in FM responses

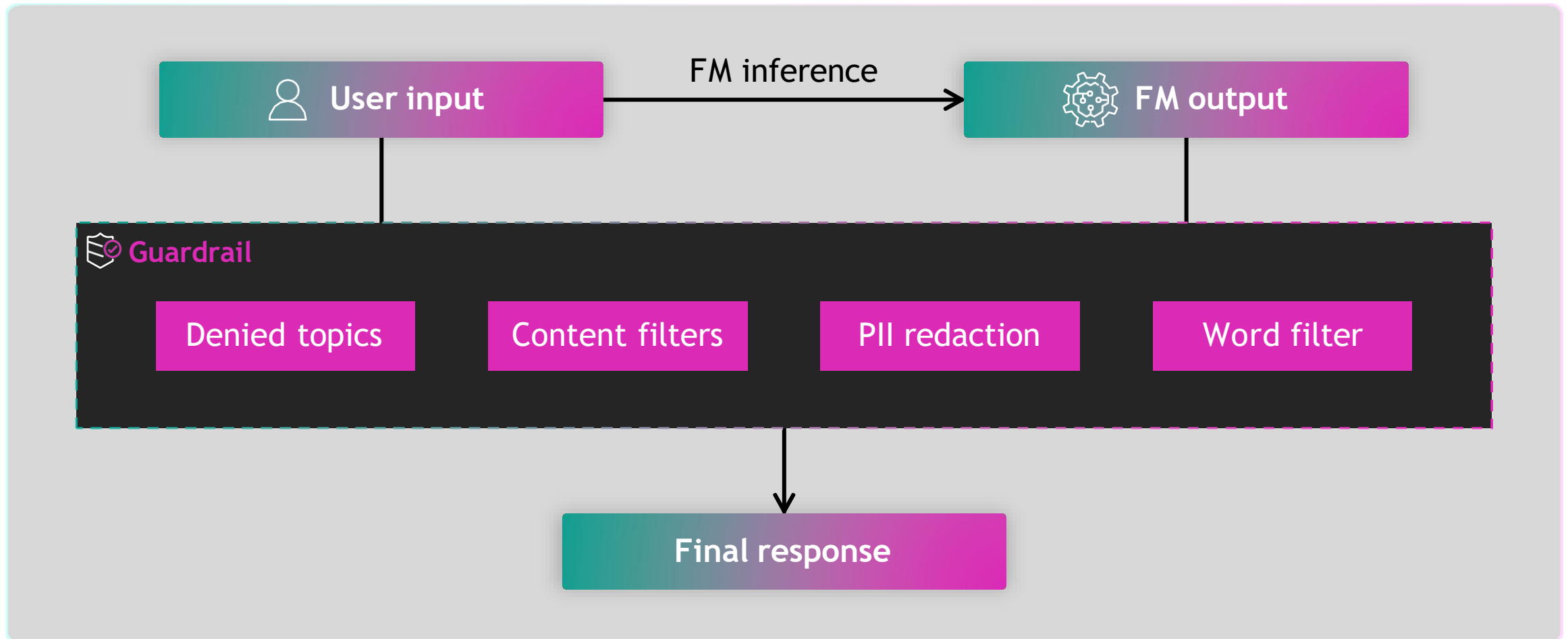
Guardrails



- Control the interaction between users and Foundation Models (FMs)
- Filter undesirable and harmful content
- Remove Personally Identifiable Information (PII)
- Enhanced privacy
- Reduce hallucinations
- Ability to create multiple Guardrails and monitor and analyze user inputs that can violate the Guardrails

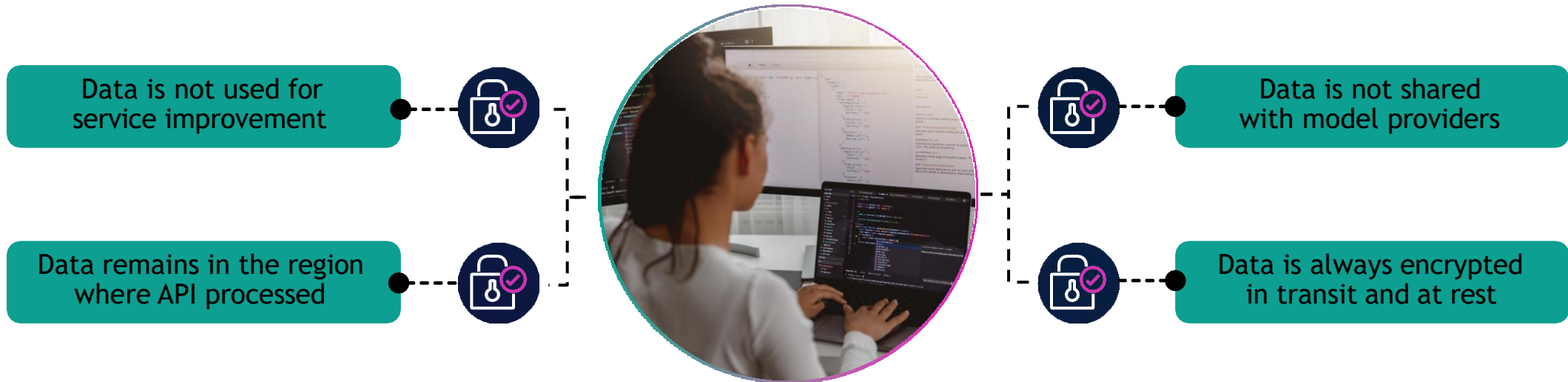


How it works



Security

Securely build generative AI applications with your data



Build with comprehensive data protection and privacy;
use AWS security services and best practices

Data Privacy and localization

You are always in control of your data

- Customer data is not used to improve any of the foundation models for other customers and is not shared with other foundation model providers
- All customer model data (prompts, responses, fine-tuned models) are isolated per customer and remain in the AWS Region where they were created

You are always in control of your data

- Customer data is always encrypted in transit with a minimum of TLS1.2 and AES-256 encrypted at rest using AWS KMS managed data encryption keys
- Fine-tuned (customized) models can be encrypted and stored using your AWS KMS keys; only you have access to your customized models
- Integration with AWS Identity and Access Management (IAM) to manage inference access, allow/deny access for specific models, enable AWS Management Console access, etc.

- **Current status:**
 - HIPAA eligible
 - GDPR, SOC compliant
 - FedRAMP Moderate - In scope
 - FedRAMP High - 3PAO Assessment

Amazon Bedrock - model subscription access

- Customers need to subscribe to a model before using it to build a generative AI application.
- IAM policies restrict access to specific models authorized by the customer for their use case.
- The policy can be fine-tuned to ensure that only specific models are available for subscription.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow|Deny",
      "Action": [
        "aws-marketplace:Subscribe|Unsubscribe|ViewSubscriptions"
      ],
      "Resource": "*",
      "Condition": {
        "ForAnyValue:StringEquals": {
          "aws-marketplace:ProductId": [
            "model-product-id-1",
            "model-product-id-2",
            "..."
          ]
        }
      }
    }
  ]
}
```

Pricing

- On-Demand
 - Pay-as-you-go (no commitment)
 - Text Models – charged for every input/output token processed
 - Embedding Models – charged for every input token processed
 - Image Models – charged for every image generated
 - Works with Base Models only
- Batch:
 - Multiple predictions at a time (output is a single file in Amazon S3)
 - Can provide discounts of up to 50%
- Provisioned Throughput
 - Purchase Model units for a certain time (1 month, 6 months...)
 - Throughput – max. number of input/output tokens processed per minute
 - Works with Base, Fine-tuned, and Custom Models

Cost savings

- On-Demand – great for unpredictable workloads, no long-term commitment
- Batch – provides up to 50% discounts
- Provisioned Throughput – (usually) not a cost-saving measure, great to “reserve” capacity
- Temperature, Top K, Top P – no impact on pricing
- Model size – usually a smaller model will be cheaper (varies based on providers)
- Number of Input and Output Tokens – main driver of cost

Thank You